Homework 1: Image Enhancement

Vittorio Pettene – VR543856
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1 Objective and images

The goal of this homework is to compare four classical enhancement methods on degraded color images, looking at the trade-off between detail visibility, noise and natural appearance. No deep learning is used: the pipeline is written in Python with OpenCV and NumPy, and scikit-image is used only for SSIM.

I used three real color photos, each with a different problem (Table 1). All images were resized to 1600 px on the long side before running the experiment.

Image	Condition	Source / License
`20190617_Nebbia_(3)`	Fog, very low contrast	Wikimedia Commons, (fill in)
`500px_photo_(65567999)`	Backlight, sun flare, non-uniform light	Wikimedia Commons, (fill in)
`At_Cueva_del_Viento_2023_069`	Cave lit by a flashlight, dark borders	Wikimedia Commons, (fill in)

Table 1: Images used in the experiment.

Important: the original images are used as the reference for the metrics, because the degradation is synthetic. In a real scenario a clean reference image is usually not available, so the metrics below only measure how close each method gets back to the original photo.
Repository (code, images, results, metrics): `https://github.com/...`

2 Degradation and methods

Degradation. Each original image I is degraded in the same way: the contrast is halved around the global mean μ, Gaussian noise is added and the result is clipped to $[0, 255]$:

$$I_{deg} = \mathrm{clip}\big(\mu + 0.5\,(I - \mu) + n\big), \qquad n \sim \mathcal{N}(0,\, 10^2). \tag{1}$$

The random seed is fixed to 42 and the images are processed in alphabetical order, so the degraded images are always the same. Since the contrast is reduced on all three channels, the colors are also weakened.

Methods. All parameters are the same for every image.

1. **Baseline:** the degraded image, without any processing.
2. **Linear contrast stretching (per channel):** each B, G, R channel is mapped linearly so that its 1st percentile goes to 0 and its 99th percentile goes to 255. Percentiles are used instead of min/max so that a few noisy pixels do not decide the range. Since each channel gets its own mapping, the colors can shift.
3. **Histogram equalization (luminance):** the image is converted to CIE-Lab and only the L channel is equalized (`cv2.equalizeHist`); a and b are not changed, so the hue should stay about the same.

4. **NL-means + stretching (combination):** non-local means denoising (`cv2.fastNlMeansDenoisingColore` $h = 10$, template 7, search window 21) followed by the same contrast stretching as method 2. The idea is to remove the noise before the stretching amplifies it. The downside is that denoising can also smooth fine details and textures.

Metrics. PSNR (dB) and SSIM (scikit-image, on the three channels) with respect to the original, plus the mean absolute error (MAE), implemented by hand. Higher PSNR and SSIM and lower MAE mean a result closer to the original.

3 Results

Table 2 reports the metrics and Figure 1 the visual comparison. NL-means + stretching has the highest SSIM on all three images, while contrast stretching and histogram equalization never improve SSIM over the baseline.

Image	Method	PSNR (dB) ↑	SSIM ↑	MAE ↓
Fog	Baseline	19.94	0.388	21.18
	Contrast stretching	17.80	0.114	26.69
	Hist. equalization (L)	13.37	0.121	47.35
	NLM + stretching	**20.38**	**0.812**	**19.54**
Backlight	Baseline	19.42	0.405	21.26
	Contrast stretching	**20.97**	0.177	**18.22**
	Hist. equalization (L)	14.54	0.184	40.57
	NLM + stretching	19.92	**0.701**	22.73
Cave	Baseline	17.01	0.511	30.99
	Contrast stretching	19.32	0.436	22.71
	Hist. equalization (L)	14.83	0.429	39.88
	NLM + stretching	**19.45**	**0.517**	**19.54**

Table 2: Metrics with respect to the original image (best value per image in bold).



Figure 1: Rows: fog, backlight, cave. Columns: original (reference), degraded input (baseline), contrast stretching, histogram equalization on L, NL-means + stretching. PSNR and SSIM are computed against the original.

4 Discussion

In general, contrast stretching and histogram equalization make the image more contrasted, but they also amplify the noise, so they never improve SSIM over the baseline. Only the method with denoising before the stretching improves SSIM, but it smooths the image.

Backlight image. Contrast stretching has the best PSNR (20.97 dB, +1.5 dB over the baseline) and the lowest MAE. However, SSIM drops from 0.405 to 0.177 and the image is visibly grainy, especially in the dark areas of the armchair and the floor. PSNR rewards the fact that the average intensity is closer to the original, while SSIM sees that the local structure is covered by noise. So here a better PSNR does not mean a better looking image. My preferred method here is NLM + stretching (SSIM 0.701): the armchair, the lamp and the curtains are clean and readable. However the result looks a bit "painted" and it is darker than the original.

Fog image. Histogram equalization has by far the lowest PSNR (13.37 dB), because it moves the histogram very far from the original foggy image. Visually it does make the buildings in the background and the roof easier to see, but it also darkens the sky and makes the noise very strong. My preferred method is again NLM + stretching, which is the best on all three metrics (20.38 dB, SSIM 0.812): the house edges on the right and the roof tiles become clearer. Its artifacts are small "watercolor" blotches in the sky and a sky pushed almost to white.

Cave image. NLM + stretching has the best PSNR and MAE, but its SSIM (0.517) is practically equal to the baseline (0.511). The noise is removed, but the fine texture of the lava floor is flattened too, and SSIM penalises this loss of detail. Stretching and equalization go below the baseline in SSIM (0.436 and 0.429).

Colors. The degradation reduces the contrast on all channels, so the colors also become weaker. Histogram equalization only changes L, so the colors stay faded and the results look a bit grey. Stretching each channel brings back some color, but since every channel has a different mapping the colors can shift and the noise becomes colored.

5 Conclusion and limitations

No method is the best on every metric and every image. With noisy, low-contrast inputs, denoising before the contrast enhancement is the most reliable choice (highest SSIM on all three images), but it smooths textures and can create blotchy artifacts. Contrast stretching and equalization show more details in dark or foggy areas but amplify the noise, so their PSNR gains can be misleading. For this reason the choice of a method is based on metrics and visual inspection together, not on a single metric.

Limitations. The degradation is synthetic and global (same contrast factor and noise level everywhere), while real images have spatially varying illumination and noise. The parameters (percentiles, $h = 10$) were fixed by hand and not tuned per image. Finally, all metrics need a clean reference, which is not available in a real application.